EEG-Based Motor Action and Motor Imagery Classification Using Machine Learning

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Abstract

This work presents an EEG-based classification framework for distinguishing motor actions and motor imagery using a consumer-grade Emotiv EPOC X headset. EEG recordings were collected from 47 participants using 14 channels sampled at 128 Hz while participants performed or imagined six limb movements involving the right and left arms, legs, and wrists. The resulting protocol formed twelve task classes by combining the body part with the action or imagery condition. The raw EDF recordings were processed using MNE-Python, including channel selection, 4–45 Hz band-pass filtering, signal-quality based trial rejection, per-recording normalisation and sliding-window segmentation. Each 10-second task segment was divided into 2-second windows with a 0.5-second stride, producing 8,092 windows from 476 trials. A heterogeneous representation of 1,771 features per window was then constructed using spectral, statistical, Hjorth, fractal and wavelet descriptors, together with spectrogram, scalogram, inter-hemispheric connectivity and Riemannian tangent-space features. An ensemble of Logistic Regression, Linear Discriminant Analysis, Extra Trees and Histogram Gradient Boosting was used as the main classifier, and Support Vector Machine, Random Forest, Naive Bayes, AdaBoost, EEGNet and CNN–LSTM models were evaluated for comparison. Subject-level GroupKFold validation was used to prevent windows from the same participant appearing in both training and evaluation folds. The ensemble achieved 80.25% for binary action-versus-imagery classification, 26.68% for six-class body-part classification and 21.43% for the full

twelve-class task, while left-versus-right imagery classification reached 52.28% under nested cross-validation. The results show that combining complementary time–frequency, covariance-based and signal-derived features provides a strong baseline for subject-independent decoding of real versus imagined movement from low-cost EEG, while limb-level and side-level decoding remain challenging.

Keywords: Electroencephalography (EEG), Brain–Computer Interface, Motor Imagery, Motor Action, Spectrogram, Scalogram, Riemannian Geometry, Machine Learning, Emotiv EPOC X

1 Introduction

Brain–computer interfaces (BCIs) give the brain a channel to a computer that does not pass through muscles. They record brain activity, look for patterns that match an intention, and turn those patterns into a command. Motor applications are where this matters most. A person who has lost control of a limb after a stroke or a spinal injury can often still produce the cortical activity that would normally drive the movement, and a BCI that reads this activity could steer a wheelchair, open a prosthetic hand or guide a rehabilitation robot. Most such systems use electroencephalography (EEG). It is non-invasive and cheap compared with other recording methods, it follows brain activity on the scale of milliseconds, and it can be recorded with equipment small enough to carry [1, 2].

Two related mental states are central to this work. In motor execution, or *action*, the person really makes the movement. In motor imagery, the same movement is rehearsed in the mind while the body stays still. The two are not the same, but they share much of their neural machinery. Both reduce the power of the sensorimotor rhythms over the motor cortex, mostly in the mu band (8–12 Hz) and the beta band (13–30 Hz), and this drop, known as event-related desynchronisation, is the signal that most motor-imagery BCIs rely on [3, 4].

Much of the early work kept the problem small. The classic task is to tell imagined left-hand movement from imagined right-hand movement, and Common Spatial Pattern (CSP) filters followed by linear discriminant analysis or a support vector machine became the standard way to solve it [5, 6]. Adding more classes makes things far harder. The cortical areas for the hand, the wrist and the leg lie close together, the skull spreads their activity across the scalp, and the EEG of one person can look quite unlike that of the next [7, 8]. A model that works well for the people it was trained on can therefore fail on someone new.

The recording hardware adds its own limits. Clinical EEG amplifiers give clean signals, but they are expensive, slow to set up and rarely leave the laboratory. Consumer headsets such as the Emotiv EPOC are cheap, quick to put on and comfortable enough for everyday use, and studies have shown that they capture real neural activity [9, 10]. They are also noisier, more sensitive to movement and limited in where they place their electrodes [11]. The EPOC X used here has fourteen saline electrodes,

none of which sits over the central motor strip (C3, Cz and C4), which is where motor activity is usually strongest.

Researchers have tried many ways of turning short EEG segments into something a classifier can use. Band-power and statistical features summarise each channel. Spectrograms and scalograms keep track of how the rhythms change over time. Riemannian methods describe a trial by the covariance between its channels and work directly with the geometry of these matrices [12, 13]. Deep networks such as EEGNet learn their own features from the raw signal [14, 15]. Each of these views captures something the others miss, so combining them is a natural step, particularly when the dataset is too small for a deep network to learn everything on its own.

How a model is evaluated matters as much as the model itself. EEG windows cut from the same recording are strongly correlated, so if windows of one person appear in both the training and the test set, the classifier can learn to recognise the person instead of the task. An early version of our own pipeline fell into exactly this trap and reported accuracies above 90%, which dropped sharply once the evaluation was corrected. For a BCI that has to work for new users, the only fair test is one in which the people being tested were never seen during training.

In this study we built a complete classification pipeline around the Emotiv EPOC X and a twelve-task protocol. Participants performed and then imagined six movements: right-arm and left-arm flexion, right-leg and left-leg lifting, and right-wrist and left-wrist rotation. We analysed the same recordings at three levels of detail: action versus imagery (two classes), the body part involved (six classes), and the exact task (twelve classes). At our guide's suggestion we also asked whether the side of an imagined movement, left or right, can be decoded. Along the way we found that the task schedule did not start at a fixed time in every recording, and we recovered the true start from the headset's own motion sensor before cutting any trials. Every result we report comes from people the model had not seen during training.

The main contributions of this work are as follows:

1. **A new twelve-task EEG dataset of real and imagined limb movements.** We recorded 47 participants with a consumer-grade 14-channel Emotiv EPOC X while each of six movements (right and left arm, leg and wrist) was first performed and then imagined.
2. **Subject-independent decoding of executed versus imagined movement from consumer-grade EEG.** We show that real and imagined movement can be told apart in people the model has never seen, at 80.25% with a 14-channel consumer headset. The result holds under person-wise cross-validation, leave-one-subject-out validation (80.25%) and nested cross-validation (79.41%).
3. **Classification of the same recordings at several levels of detail.** Every trial is classified in three ways: action or imagery (2 classes), the body part that was moved or imagined (6 classes), and the exact task, meaning body part and condition together (12 classes). In addition, an imagery-versus-imagery binary task uses only the imagery trials and asks whether the imagined movement was on the left or the right side of the body. Because all of these use the same trials, they show directly how accuracy falls as the question becomes more specific: 80.25% for action

versus imagery, 26.68% for body part (chance 16.67%), 21.43% for the twelve tasks (chance 8.33%), and 52.28% for the imagery-versus-imagery (left versus right) task.

4. **A controlled comparison of feature representations and classifiers under one subject-independent protocol.** On identical folds we compare handcrafted, spectrogram, scalogram, connectivity, CSP and Riemannian features, and classical, ensemble and deep-learning models. The Morlet scalogram is consistently more informative than the STFT spectrogram, adding CSP features to each classifier changes accuracy by less than two points, and deep networks (EEGNet, CNN–LSTM) do not outperform feature-based models on a dataset of this size.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the protocol, data acquisition, preprocessing, feature extraction, classifiers and evaluation. Section 4 presents and discusses the results, and Section 5 concludes the paper.

2 Literature Review

Decoding movement from scalp EEG has been studied for several decades, and the main obstacle has always been the same: the signal is weak, noisy and different from one person to the next. The physiological basis of almost all motor decoding is a change in the sensorimotor rhythms. When a person moves, or prepares to move, the power of the mu rhythm (8–12 Hz) and the beta rhythm (13–30 Hz) over the motor cortex falls, and it rises again, often above its resting level, once the movement ends. These two effects are called event-related desynchronisation and synchronisation (ERD/ERS) [4, 16]. Neuper and Pfurtscheller showed that their frequency, timing and location depend on the body part involved, which is what makes it possible, in principle, to tell one movement from another [17].

Motor imagery produces a similar pattern without any overt movement. Studies of the neural basis of imagery have found that imagining an action engages much of the same motor network as performing it, although usually more weakly [18]. This overlap is why imagery can drive a BCI, but it also means that action and imagery are not trivially separable. Pfurtscheller et al. found that imagined hand, foot and tongue movements each change the mu rhythm in their own characteristic way, and that these differences can be used to classify single trials, although not equally well for every person [19]. Most BCI work has concentrated on imagery alone. Distinguishing real from imagined movement, which matters for rehabilitation systems that must know whether a patient is trying to move or only thinking about it, has received far less attention.

Common Spatial Pattern (CSP) filtering is probably the most influential single method in the field. Ramoser et al. showed that spatial filters which maximise the difference in variance between two classes give strong results for imagined left- and right-hand movement [5]. Blankertz et al. later explained why plain CSP is fragile when the data are noisy or the training set is small, and how regularised variants make it more stable [6]. Filter-bank CSP (FBCSP) added the observation that the most useful frequency band differs from person to person, so it runs CSP in several sub-bands and keeps the most informative features [20]. FBCSP won the four-class and

two-class motor imagery tasks of BCI Competition IV and is still one of the standard baselines [21].

Covariance-based methods grew out of the limits of spatial filtering. Barachant et al. treated the covariance matrix of each trial as a point on the manifold of symmetric positive-definite matrices and classified trials by their Riemannian distance to the mean of each class [12]. Projecting the matrices onto a tangent space turns them into ordinary vectors that standard classifiers can use, while keeping the relationships between channels. Congedo et al. reviewed this family of methods and argued that they are both simpler and more robust than many earlier pipelines, especially for multi-class problems [13].

Time–frequency analysis looks at the same signals from another angle. ERD/ERS unfolds over time, so averaging power over a whole window can hide part of it. The short-time Fourier transform (STFT) produces a spectrogram that shows how band power changes during a trial, and Tabar and Halici used STFT images of motor-imagery EEG as the input to a convolutional network [22]. The continuous wavelet transform (CWT) produces a scalogram instead. It analyses high frequencies with short windows and low frequencies with long ones, which suits signals whose rhythms come and go [23]. Simpler descriptors remain widely used as well: band power in fixed frequency bands [24], the Hjorth parameters, which summarise the activity, mobility and complexity of a signal [25], and fractal measures of how irregular the signal is [26].

Deep learning is a further major direction. Schirrmeister et al. showed that convolutional networks trained on raw EEG can match FBCSP on motor imagery, and visualised what the networks had learned [14]. Sakhavi et al. fed a temporal representation of filter-bank envelopes into a CNN to capture how the signal evolves within a trial [27]. EEGNet showed that a compact network built from depthwise and separable convolutions can work across several BCI paradigms with very few parameters [15]. Reviews of deep learning for EEG, however, note that its advantage over well-designed classical pipelines is often small and uneven, particularly when the number of participants is limited [28–30].

Generalising to people who were not in the training set is still one of the hardest problems. The EEG of different people differs in amplitude, in the exact position of the electrodes and in how each person carries out the task, so a classifier trained on one group often performs poorly on another [8]. Alignment methods reduce this shift by re-centring each person's data before classification, either in Euclidean space [31] or on the Riemannian manifold [32, 33]. The MOABB benchmark compared many algorithms on many public datasets and showed how strongly the ranking of methods can change from one dataset to the next, which argues for testing under realistic, subject-independent conditions rather than reporting a single favourable split [34].

Consumer-grade EEG lowers the cost of all this but adds problems of its own. Badcock et al. compared the Emotiv EPOC with a research system and found that it captured auditory event-related potentials of usable quality [9]. Duvinage et al. found it workable for a P300 speller, though clearly less accurate than a medical-grade amplifier [10], and Maskeliūnas et al. reported high variability between trials when consumer headsets were used for control tasks [11]. Motor-imagery BCIs have

nevertheless reached clinical use; Ang et al., for example, combined CSP-based imagery detection with robotic feedback in a randomised trial of stroke rehabilitation [35].

Taken together, these studies leave a clear gap. Most motor-imagery work uses research-grade amplifiers, public benchmark datasets and two or four imagery classes. Far fewer studies separate real from imagined movement, compare several levels of detail on the same recordings, or test consumer hardware under strictly subject-independent evaluation. Our study addresses this gap. Table 1 summarises the representative studies discussed above.

Table 1 Comparison of representative studies on EEG-based motor imagery and motor-related classification. N/R: not reported in a directly comparable form.

Ref.	Task / Classes	Dataset / Hardware	Features / Representation	Model	Accuracy	Main Limitation
[5]	Left vs. right hand MI (2-class)	Research EEG, 3 subjects	CSP + log-variance	LDA	≈80–90%	Restricted to binary MI
[19]	Hand, foot, tongue MI	Research EEG	Mu-band ERD/ERS patterns	Linear classifier	N/R	Imagery only; varies by person
[7]	Hand, foot, tongue MI (4-class)	Research EEG	ICA-based representation	LDA	N/R	Limited spatial separability
[20]	Motor imagery (4-class)	BCI Competition IV-2a	Filter-bank CSP + feature selection	Naive Bayes	≈68%	Many frequency/filter parameters
[12]	Multi-class MI	Research EEG	Covariance matrices + Riemannian geometry	MDM / tangent-space classifier	N/R	Costly covariance operations
[22]	Motor imagery (2-class)	BCI Competition IV-2b	STFT images	CNN + stacked autoencoder	≈77%	Complex, data-sensitive training
[14]	Motor imagery (4-class)	BCI Competition IV-2a	Learned spatio-temporal features	Deep / Shallow ConvNet	≈71–74%	Data-hungry, less interpretable
[27]	Motor imagery (4-class)	BCI Competition IV-2a	Temporal filter-bank representation	CNN	≈74.5%	More complex architecture
[15]	Several BCI paradigms	Public EEG datasets	Depthwise and separable convolutions	EEGNet	N/R	Needs enough training data
[31]	Cross-subject MI/ERP	Public EEG datasets	Euclidean alignment of covariance	Various classifiers	N/R	Assumes a moderate covariance shift
[32]	Cross-subject MI	Public EEG datasets	Riemannian re-centring of covariances	MDM	N/R	Needs unlabelled data of the new subject
[33]	Cross-subject EEG	Four EEG datasets	Riemannian alignment + tangent space	MEKT	N/R	Needs source/-target adaptation
[9]	EEG signal validation	Emotiv EPOC (14-ch)	Auditory event-related potentials	ERP analysis	N/R	Lower spatial resolution than clinical EEG
[11]	Concentration / blink control	Emotiv EPOC, NeuroSky	EEG control features	Conventional classifiers	N/R	High inter-trial variability
[35]	MI-based stroke rehabilitation	Stroke patients	CSP-based MI features	BCI + robotic feedback	N/R	Small clinical cohort
Our work	Action vs. imagery (2), body part (6), twelve tasks (12), imagery vs. imagery left/right (2)	Emotiv EPOC X (14-ch, 128 Hz), 47 participants	1,771 features: handcrafted + STFT + CWT + connectivity + Riemannian	Ensemble (LR + LDA + Extra Trees + HistGB), mirror augmentation, trial-level voting	80.25% (2-class); 26.68% (6-class); 21.43% (12-class); 52.28% (imagery L/R, nested CV)	Consumer EEG with no central electrodes; limb and side remain hard to decode

3 Proposed Methodology

3.1 System Overview

The system takes the raw files produced by the headset and returns one label for every 10 s task. It works in two stages (Fig. 1). The first stage, *signal preparation*, turns a continuous recording into clean, labelled trials: it finds where the task schedule starts, filters the EEG, cuts out the twelve trials, removes the ones spoiled by artefacts and normalises what is left. The second stage, *representation and classification*, splits each trial into short overlapping windows, describes every window with spectral, time–frequency, connectivity and covariance features, aligns these features within each recording and classifies the window with an ensemble of four models. The predictions for the windows of a trial are then averaged into one decision. Throughout the pipeline, no participant's data are ever used for both training and testing.

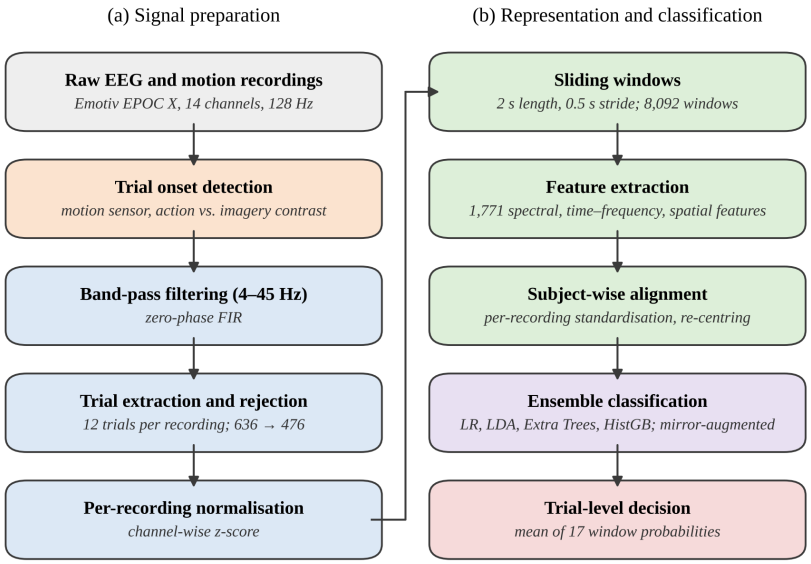


Fig. 1 Overview of the proposed pipeline.

3.2 Experimental Protocol

Participants sat in front of a screen in a quiet room. They were asked to keep their eyes open, to follow the instruction shown on the screen, to avoid unnecessary movement and speech, and to relax whenever a fixation cross appeared. The protocol covers six movements: right-arm flexion, left-arm flexion, right-leg lifting, left-leg lifting, right-wrist rotation and left-wrist rotation. Each movement was first carried out physically (*action*) and then rehearsed mentally without moving (*imagery*), giving twelve tasks in a fixed order:

`RightArm_Action`	`RightArm_Imagine`
`LeftArm_Action`	`LeftArm_Imagine`
`RightLeg_Action`	`RightLeg_Imagine`
`LeftLeg_Action`	`LeftLeg_Imagine`
`RightWrist_Action`	`RightWrist_Imagine`
`LeftWrist_Action`	`LeftWrist_Imagine`

Each task lasted 10 s and was followed by 5 s of rest, so the whole schedule took 175 s from the start of the first task to the end of the last. Two session layouts occur in the data (Fig. 2). In most sessions the tasks began a few seconds after the recording was started. In several later sessions the recording opened with a resting baseline, about 50 s with eyes open followed by about 50 s with eyes closed, so the tasks began roughly two minutes in. Every trial carries three labels: the task itself (twelve classes), the condition, action or imagery (two classes), and the body part (six classes).

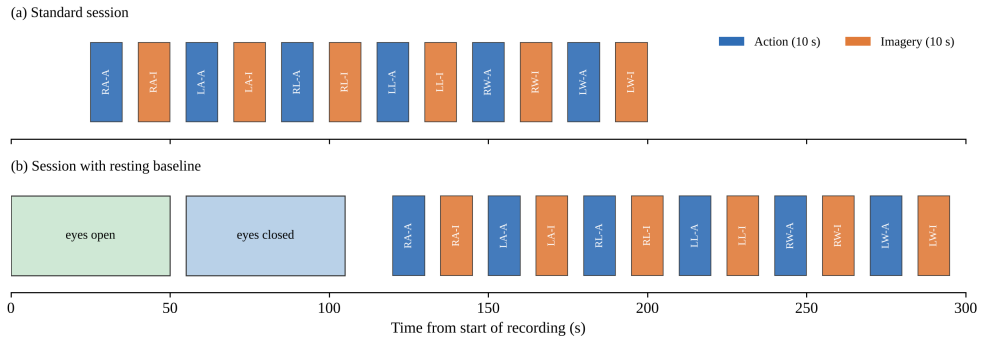


Fig. 2 Timeline of a recording session: (a) standard session; (b) session with a resting baseline.

3.3 Data Acquisition

EEG was recorded with a 14-channel Emotiv EPOC X fitted with saline-soaked felt electrodes at AF3, AF4, F3, F4, F7, F8, FC5, FC6, T7, T8, P7, P8, O1 and O2, placed according to the international 10–20 system, with the headset's CMS/DRL reference. The signals were sampled at 128 Hz and saved in EDF format by the EmotivPRO software. For every session EmotivPRO also saves a second file from the headset's built-in inertial sensor, which holds three accelerometer axes and four quaternion (orientation) components sampled at 32 Hz. We never used this motion file as a classification input; it served only to locate the task schedule (Section 3.4). In total we collected 59 EEG recordings containing all fourteen channels from 47 participants. A few participants were recorded in more than one session.

3.4 Motion-Guided Trial Onset Detection

To cut the trials correctly, we need to know when the first task begins in each recording. Our first pipeline assumed a fixed delay of 15 s for every file. When we checked this against the data it turned out to be wrong: the delay between starting the recording and starting the tasks changed from session to session, and in sessions with a baseline it was more than two minutes. With a fixed offset, most trials were cut several seconds early, and a window labelled with one task often contained the end of the previous one.

To recover the real onset we used a simple physical fact. Moving an arm, a leg or a wrist shakes the head slightly, and the headset's motion sensor picks this up; imagining the same movement does not. The procedure has four steps.

Step 1: motion energy. The motion signal is divided into 0.5 s bins. For each bin b, the motion energy is the logarithm of the total absolute change across the seven inertial channels,

$$m_b = \log \sum_{c=1}^{7} \sum_{n \in b} \left| s_c[n] - s_c[n-1] \right|, \tag{1}$$

and the sequence m_b is centred on its median and divided by its standard deviation.

Step 2: scoring each candidate onset. For every candidate onset t_0, in steps of 0.5 s, the twelve-task schedule is laid over the motion signal. We take the mean motion energy over seconds 1–10 of each task, skipping the first second, when the participant is still reacting to the cue. For each of the six action–imagery pairs we compute the difference d_k, and from these a paired t-statistic:

$$d_k = \bar{m}_{\text{action},k} - \bar{m}_{\text{imagery},k}, \qquad t(t_0) = \frac{\bar{d}}{s_d/\sqrt{6}}, \qquad k = 1, \ldots, 6. \tag{2}$$

Step 3: choosing the onset. The candidate with the largest t is selected. We treat it as *confident* if $t \geq 4$ and all six pairs show more motion during action than during imagery.

Step 4: an independent check. Recordings were normally stopped soon after the last task, and across the confident recordings the file ended a median of 181 s after the first task began. This gives a second, label-free estimate of the onset, $\hat{t}_0 = T_{\text{rec}} - 181$, where T_{rec} is the length of the recording. An onset that fails Step 3 is still accepted if it lies within 5 s of this estimate. Otherwise the recording is dropped, since mislabelled trials would do more harm than a slightly smaller dataset.

Fig. 3 shows the procedure for one recording. Motion energy rises clearly in every action block and stays low in the imagery blocks. The t-statistic peaks sharply at the detected onset of 26.0 s ($t = 24.8$), whereas at the old fixed onset of 15 s it is slightly negative, which means that under the old timing the "action" windows of this recording contained *less* movement than the "imagery" windows.

Across the dataset, 48 of the 59 recordings gave a confident onset and 5 more passed the duration check. Two were excluded because the two estimates disagreed, and four were too short to contain a complete schedule (sessions that had been stopped and restarted). The detected onsets ranged from 7.5 s to 136.5 s, and the median gap between the motion-based and the duration-based estimates was 2.5 s. In the seven

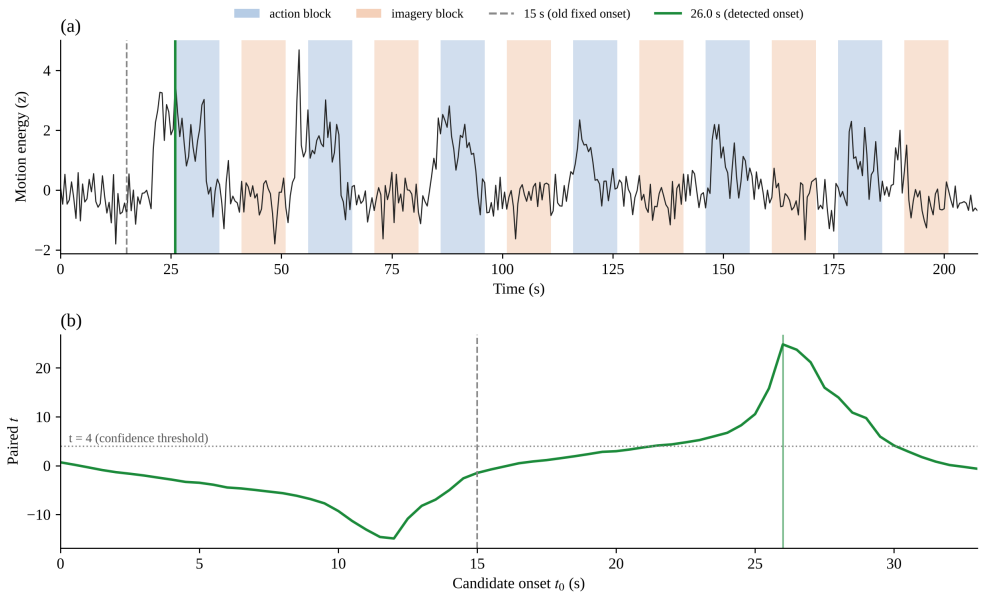


Fig. 3 Motion-guided onset detection for one recording: (a) head-motion energy with the detected schedule; (b) paired t-statistic for every candidate onset.

sessions where the eyes-closed baseline could be confirmed by a clear rise in occipital alpha power, the detected onset fell 7.3–15.5 s after the end of the eyes-closed period, which is where the schedule was expected to begin. The remaining 53 recordings were used in all later steps.

3.5 EEG Preprocessing

Preprocessing has three parts: filtering, trial rejection and normalisation (Fig. 4).

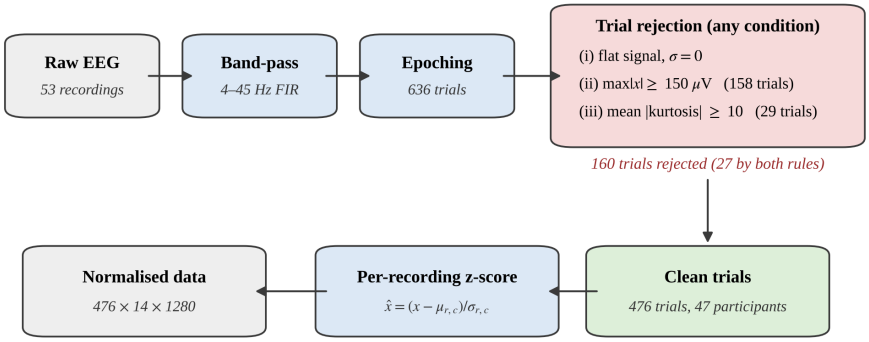


Fig. 4 Filtering, trial rejection and per-recording normalisation.

Band-pass filtering. The EDF files were loaded with MNE-Python [36], and only the fourteen EEG channels were kept; the quality, battery and marker channels were discarded and missing samples were set to zero. Each continuous recording was filtered between 4 and 45 Hz with a zero-phase FIR filter (Hamming-window design, MNE defaults). The 4 Hz edge removes slow electrode drift and most of the energy of blinks and eye movements. The 45 Hz edge removes high-frequency muscle activity and the 50 Hz mains interference. What remains covers the theta, mu, beta and low-gamma rhythms. Because the filter is zero-phase it does not shift the signal in time, and filtering the whole recording before cutting the trials avoids distortion at the trial edges.

Trial extraction. Starting from the detected onset t_0, trial k ($k = 0, \ldots, 11$) runs from $t_0 + 15k$ to $t_0 + 15k + 10$ s. Each trial is therefore a 14×1280 matrix, and the 53 recordings give 636 trials.

Trial rejection. A trial was discarded if it failed any of three checks. (i) It was flat, with zero variance over the whole trial. (ii) Any sample reached 150 μV. Genuine EEG on this device stays well below this level (the median peak of a trial was 92 μV), and larger values nearly always come from blinks, jaw clenching or a moving cable. (iii) The mean absolute excess kurtosis across channels was 10 or more, which flags sudden spikes and electrode pops that stay under the amplitude limit. No trial was flat. The amplitude rule caught 158 trials and the kurtosis rule 29, with 27 caught by both, so 160 trials (25.2%) were removed. This left **476 clean trials** from 51 recordings and 47 participants, a data array of size $476 \times 14 \times 1280$. Rejection was done before windowing, so that a noisy trial could not enter the training data as many overlapping windows. The classes stayed balanced: 235 action and 241 imagery trials, and between 36 and 44 trials for each of the twelve tasks.

Per-recording normalisation. Signal amplitude changes from session to session with electrode impedance, the amount of saline and head shape. We therefore z-scored every channel c of every recording r separately,

$$\hat{x}_{r,c}[n] = \frac{x_{r,c}[n] - \mu_{r,c}}{\sigma_{r,c}}, \tag{3}$$

where $\mu_{r,c}$ and $\sigma_{r,c}$ are the mean and standard deviation of that channel over all retained trials of the recording. These statistics pool all twelve tasks, so no class labels are involved.

3.6 Sliding-Window Segmentation

A 10 s trial is long compared with the short bursts of mu and beta change that accompany a movement, and 476 trials are not many for training a classifier. Each trial was therefore split into overlapping 2 s windows with a stride of 0.5 s (Fig. 5). At 128 Hz a window holds $2 \times 128 = 256$ samples, neighbouring windows overlap by 75%, and one trial yields

$$N_w = \frac{10 - 2}{0.5} + 1 = 17 \tag{4}$$

windows. The 476 trials thus give $476 \times 17 = 8{,}092$ windows, an input array of size $8092 \times 14 \times 256$. Every window keeps the labels of its trial, together with the index of

the trial and the identity of the participant. The trial index is used later to combine the 17 window predictions into one decision, and the participant identity keeps all windows of a person in the same cross-validation fold.

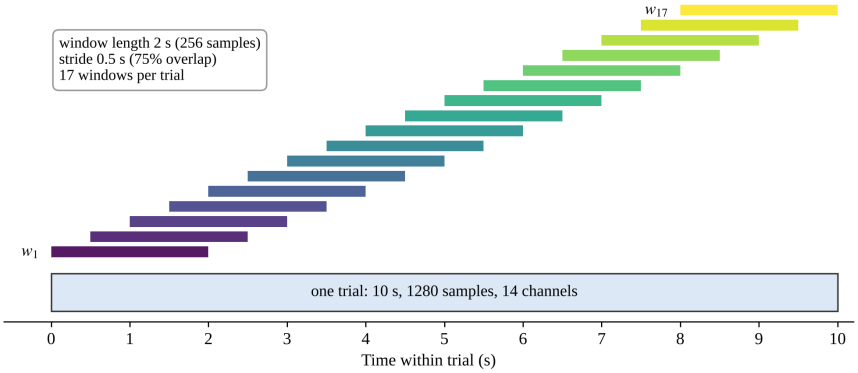


Fig. 5 Sliding-window segmentation of one trial.

3.7 Feature Extraction

Each 2 s window is turned into a fixed-length vector built from five families of features (Fig. 6): handcrafted spectral, statistical, fractal and wavelet measures; a spectrogram; a scalogram; inter-hemispheric connectivity; and the spatial covariance of the fourteen channels. The first four families give 1,666 values and the covariance family gives 105, for 1,771 features per window. None of them uses class labels.

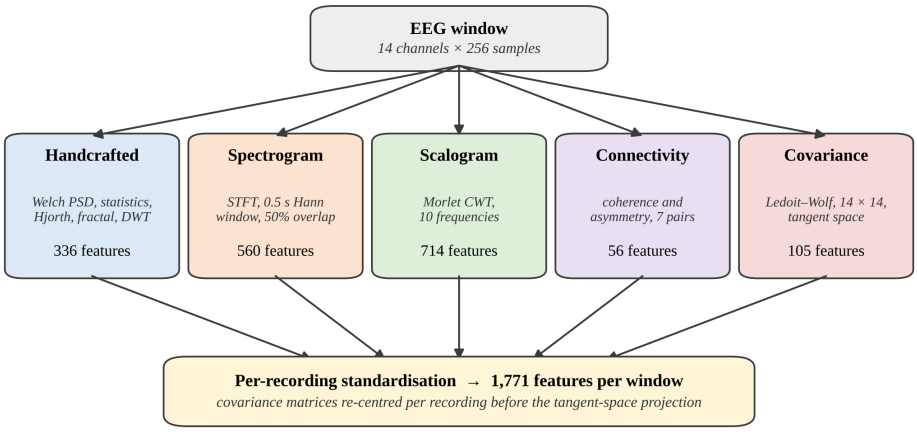


Fig. 6 The five feature families computed for every window.

3.7.1 Handcrafted Features

Twenty-four features are computed for each of the 14 channels, which gives 336 features. The power spectral density (PSD) of each channel is estimated with Welch's method (1 s Hann segments, 50% overlap, 1 Hz resolution), and the mean power is taken in four bands: theta (4–8 Hz), mu (8–12 Hz), beta (13–30 Hz) and gamma (30–45 Hz). For band b with frequency bins F_b,

$$P_b = \frac{1}{|F_b|} \sum_{f \in F_b} \mathrm{PSD}(f). \tag{5}$$

Each band is kept on a log scale, $\log(1 + P_b)$, and as relative power, $P_b / \overline{\mathrm{PSD}}$. Three ratios are added: beta to mu, mu to theta, and a sensorimotor ratio $(P_\mu + P_\beta)/(P_\theta + P_\gamma)$. The time-domain group contains the variance, kurtosis, skewness and mean absolute value of the signal, together with the Hjorth mobility and complexity [25],

$$\mathrm{Mobility} = \sqrt{\frac{\mathrm{var}(x')}{\mathrm{var}(x)}}, \qquad \mathrm{Complexity} = \frac{\mathrm{Mobility}(x')}{\mathrm{Mobility}(x)}, \tag{6}$$

where x' is the first difference of the signal. Three non-linear measures describe how irregular the signal is: the Higuchi fractal dimension with $k_{\max} = 8$ [26], the Petrosian fractal dimension, and the Hurst exponent estimated by rescaled-range analysis. Finally, a three-level discrete wavelet transform with the Daubechies-4 wavelet splits the window into bands of roughly 0–8, 8–16, 16–32 and 32–64 Hz, and the log energy of each band is kept.

3.7.2 Spectrogram Features (STFT)

Band power over a whole window tells us how much of a rhythm is present, but not when. Event-related desynchronisation, however, is a change over time. To keep this information we compute a spectrogram with the short-time Fourier transform,

$$S(t, f) = \left| \sum_{n} x[n]\, w[n - t]\, e^{-j 2\pi f n / N} \right|^2, \tag{7}$$

where w is a 0.5 s (64-sample) Hann window moved in 0.25 s steps. This gives 9 time frames per window and a frequency resolution of 2 Hz. For each channel and band, the power in each frame is averaged over the band and log-transformed (9 values), and the variance across frames is added as a measure of how unsteady the band power is. The result is $14 \times 4 \times 10 = 560$ spectrogram features.

Fig. 7 shows the mean spectrogram of all action windows and all imagery windows at four channels, together with their difference. At all four sites, imagery windows carry about 0.8–2 dB more power in the 8–13 Hz mu band than action windows, which fits the stronger mu suppression expected during real movement. Action windows, in turn, show more power below 8 Hz at the frontal channel AF3 and above 30 Hz at

FC5 and T8. This points to eye and head movement and to muscle activity while the participant moves.

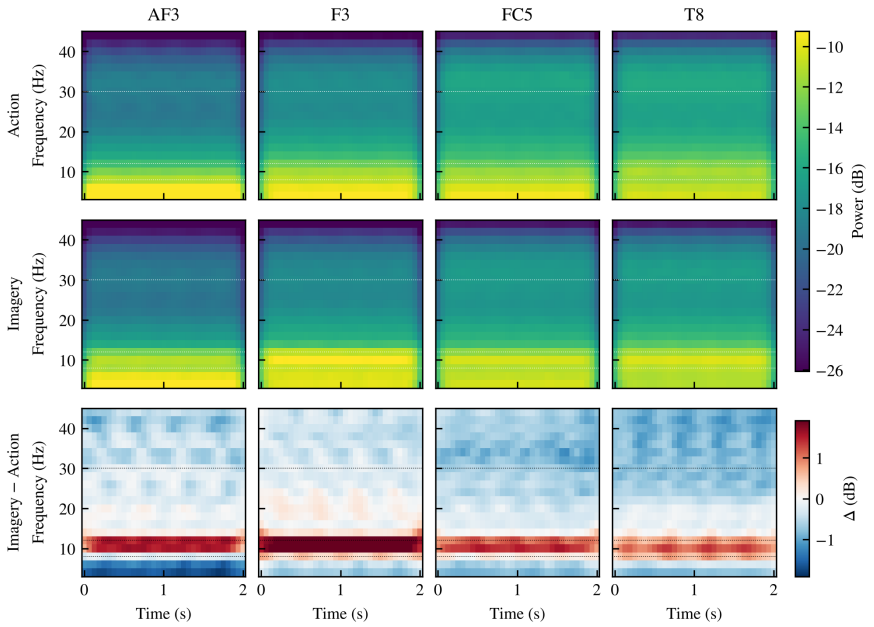


Fig. 7 Mean STFT spectrograms of action and imagery windows at AF3, F3, FC5 and T8, and their difference (imagery minus action).

3.7.3 Scalogram Features (CWT)

The STFT uses the same window at every frequency, so it has to trade time resolution against frequency resolution. The continuous wavelet transform avoids this by stretching the analysing wavelet with frequency, short for fast rhythms and long for slow ones [23]. We use the real-valued Morlet wavelet ψ:

$$W(a,\tau) = \frac{1}{\sqrt{a}} \int x(t)\, \psi\!\left(\frac{t-\tau}{a}\right) dt, \qquad \text{Scalogram}(a,\tau) = |W(a,\tau)|^2, \tag{8}$$

with scales a chosen so that the centre frequencies are exactly 5, 7, 10, 12, 15, 20, 25, 30, 35 and 40 Hz. Three groups of values are kept for each channel: the log energy at each frequency in four 0.5 s segments (40 values), the ratio of high-frequency (20–40 Hz) to low-frequency (5–15 Hz) energy (1 value), and the log variance over time at each frequency (10 values). This gives $14 \times 51 = 714$ scalogram features. The mean scalograms (Fig. 8) show the same pattern as the spectrograms, with more mu power during imagery, but sharper in both time and frequency, particularly around 10–12 Hz.

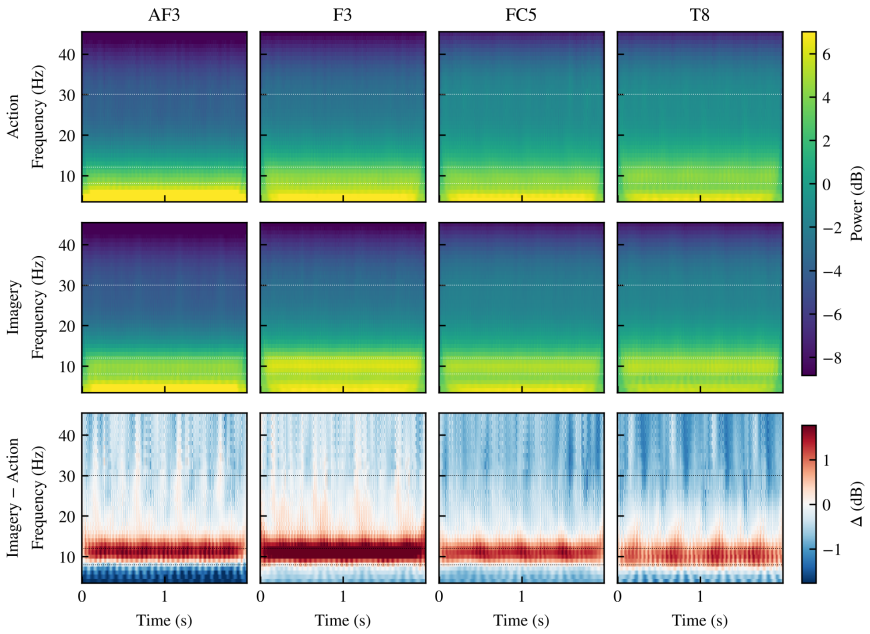


Fig. 8 Mean Morlet scalograms of action and imagery windows and their difference, for the channels of Fig. 7.

3.7.4 Connectivity and Riemannian Covariance Features

Inter-hemispheric connectivity (56 features). Moving one side of the body mainly engages the opposite hemisphere. The fourteen electrodes form seven left–right pairs (AF3–AF4, F3–F4, F7–F8, FC5–FC6, T7–T8, P7–P8 and O1–O2). For each pair and each of the four bands we compute the magnitude-squared coherence and the power asymmetry $\log(P_{\text{left}}/P_{\text{right}})$, which gives $7 \times 4 \times 2 = 56$ features.

Riemannian covariance features (105 features). For each window $X \in \mathbb{R}^{14\times 256}$ we estimate the spatial covariance matrix

$$C = \frac{1}{T-1} X X^{\top} \tag{9}$$

with Ledoit–Wolf shrinkage [37], which keeps the matrix well conditioned with only $T = 256$ samples. Covariance matrices are symmetric and positive definite, and they lie on a curved Riemannian manifold rather than in an ordinary vector space [12]. To remove the offset between sessions, the matrices of each recording r are re-centred on their own Riemannian mean M_r [32],

$$\tilde{C} = M_r^{-1/2}\, C\, M_r^{-1/2}, \tag{10}$$

so that every recording is centred on the identity matrix. The re-centred matrices are projected onto the tangent space at the identity, and the upper triangle is kept as a

vector of $14 \times 15/2 = 105$ values. M_r is computed from all windows of the recording, without labels.

3.8 Subject-Wise Feature Alignment

Normalising the raw signal is not enough on its own: feature values still sit in different ranges for different people, and one person's mu power may simply be higher than another's. Each of the 1,666 non-covariance features is therefore standardised separately for every recording r, using that recording's own windows,

$$\hat{z}_{r,j} = \frac{z_{r,j} - \mu_{r,j}}{\sigma_{r,j}}, \qquad j = 1, \ldots, 1666. \tag{11}$$

Together with the covariance re-centring, this places every recording around the same reference point. The statistics pool all twelve tasks and use no labels, so in practice a new user would only need to provide one unlabelled calibration session.

3.9 Hemispheric Mirror Augmentation

The electrodes of the EPOC X are arranged in symmetric pairs. Since a limb on one side of the body is controlled mainly by the opposite hemisphere, swapping each left electrode with its right partner gives a rough picture of the same movement made on the other side. We use this to add a mirrored copy of every training window (Fig. 9). Because every feature is computed per channel or per channel pair, the mirrored features can be built directly. The handcrafted, spectrogram and scalogram columns of partner electrodes are swapped, coherence stays the same, the sign of the asymmetry is flipped, and the re-centred covariance matrix is permuted, $\tilde{C}' = P\tilde{C}P^{\top}$, before it is projected to the tangent space. The labels are mirrored in the same way (right arm becomes left arm, right leg becomes left leg, right wrist becomes left wrist), while the action or imagery condition is kept. Mirrored copies are added to the training folds only; test windows are never altered.

3.10 Classification Models

Proposed ensemble. The 1,771 aligned features are scaled with a standard scaler fitted on the training participants and then passed to four classifiers of different types (Table 2). Each classifier outputs class probabilities for a window w, and the ensemble averages them with equal weights,

$$p_w = \frac{1}{4} \sum_{m=1}^{4} p_w^{(m)}. \tag{12}$$

Linear models and tree ensembles tend to make different mistakes, so averaging them evens out part of the error. The four models and their settings were fixed in advance and used without change for every task.

Comparison models. To put the ensemble in context, we evaluated a set of comparison models on the same data and folds, at window level and without subject-wise

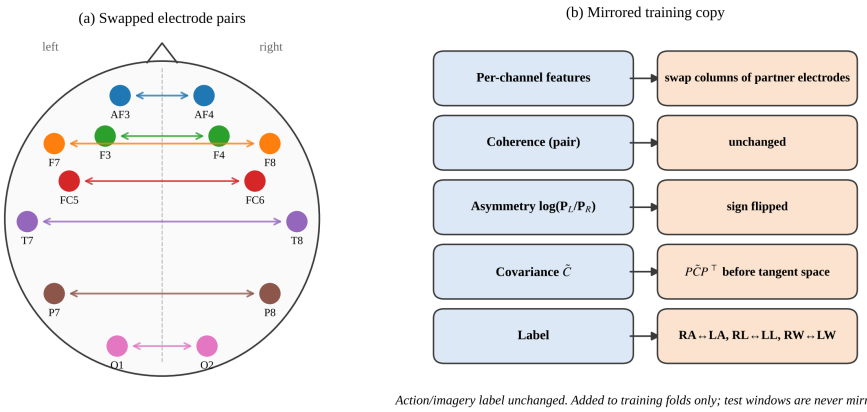


Fig. 9 Hemispheric mirror augmentation: (a) electrode pairs that are swapped; (b) how each feature group and the label are mirrored.

Table 2 Classifiers of the proposed ensemble.

Model	Setting	Reason for inclusion
Logistic Regression (LR)	L2 penalty, $C = 0.01$	Strongly regularised linear model
Linear Discriminant Analysis (LDA)	Least squares, automatic Ledoit–Wolf shrinkage	Standard EEG classifier; copes with many features
Extra Trees	500 trees, $\sqrt{d}$ features per split	Non-linear, low-variance averaging of random trees
Histogram Gradient Boosting (HistGB)	200 iterations, learning rate 0.05, depth 4	Non-linear, corrects its errors step by step

alignment. These were a support vector machine (RBF kernel, $C = 10$) [38], a random forest (300 trees, maximum depth 15) [39], histogram gradient boosting, Gaussian naive Bayes, AdaBoost (100 estimators) and shrinkage LDA. Each was applied to five feature sets: filter-bank CSP with four components in each of seven sub-bands between 4 and 40 Hz [20]; all 1,666 features; the top-k features ranked by ANOVA F-score ($k = 150$ for the binary task and 200 for the multi-class tasks); the Riemannian tangent-space features; and a combined set of CSP, all features and Riemannian features. Naive Bayes and AdaBoost were left out of the multi-class runs because of their run time. We also tested Riemannian minimum distance to mean (MDM) [12], and a hierarchical model that first predicts action or imagery and then the body part. In every comparison model the scaler, the feature selection, the CSP filters and the tangent-space reference were fitted inside each fold on the training participants only.

Deep-learning models. Two neural networks were trained directly on the raw 14×256 windows. EEGNet [15] used 8 temporal filters with a 64-sample kernel, a depth multiplier of 2, 16 separable filters and a dropout rate of 0.5. The CNN–LSTM used two one-dimensional convolution blocks (32 and 64 filters, each with batch normalisation, pooling and dropout), followed by two LSTM layers of 64 and 32 units [40] and a small dense layer. Both were trained with the Adam optimiser and a batch size of 64. Early

stopping was based on a validation set made up of 15% of the *training* participants, never the test participants.

CSP added to every classifier. Because CSP is the classic feature for motor imagery, we also asked whether it improves our own pipeline. In every fold, CSP filters with Ledoit–Wolf regularisation (four components in each of the seven sub-bands) were fitted on the training windows and their mirrored copies. The resulting 28 log-variance features were standardised within each recording and appended to the 1,771 features. Eight classifiers (LR, LDA, SVM, random forest, extra trees, HistGB, AdaBoost and naive Bayes) and the ensemble were then trained with and without these extra features, all with alignment, mirror augmentation and trial averaging.

3.11 Imagery-versus-Imagery (Left versus Right) Task

The imagery-versus-imagery binary task asks whether the side of an imagined movement can be decoded. Only the 241 imagery trials are used, and the target is the side of the body, left or right, pooled over arm, leg and wrist. For this task the per-recording standardisation of Section 3.8 is computed from that recording's imagery windows only (three left and three right trials) rather than from all twelve tasks; the side labels are not used. Mirror augmentation is applied as before, with the side label flipped. As a reference we also ran the standard method for this kind of problem, filter-bank CSP followed by shrinkage LDA [20], and all eight classifiers listed above. Significance against the 50% chance level was tested with a one-sided binomial test.

3.12 Trial-Level Decision and Evaluation

Trial-level decision. A working BCI makes one decision per task, not one per 2 s window. The 17 window probabilities of a trial are therefore averaged, and the class with the highest mean probability is chosen (Fig. 10c):

$$\bar{p} = \frac{1}{17} \sum_{w=1}^{17} p_w, \qquad \hat{y} = \arg\max_c \bar{p}_c. \tag{13}$$

All windows of a trial always belong to the same test fold, so this averaging never mixes training and test data. Window-level accuracy is reported as well.

Person-wise GroupKFold. The main results use GroupKFold cross-validation grouped by participant (Fig. 10a). The participants are divided into folds, and each of them is tested exactly once, by a model that never saw any of their data. Five folds were used for the two binary tasks (9–10 test participants per fold) and three folds for the six- and twelve-class tasks (15–16 test participants per fold), so that every class keeps enough training examples. Folds are grouped by person rather than by recording, so a participant with two sessions can never appear on both sides of a split. The neural networks and the hierarchical model used three folds for all tasks.

Leave-one-subject-out (LOSO). As a stricter check, the ensemble and its four members were also evaluated with leave-one-subject-out validation (Fig. 10b). Each of the 47 participants is held out in turn, and a new model is trained on the other 46, giving 47 folds per task. The label-free steps are unchanged, and the scaler, the classifiers and

mirror augmentation use the 46 training participants only. For the imagery-versus-imagery task, SVM, random forest, AdaBoost and naive Bayes were evaluated with LOSO as well.

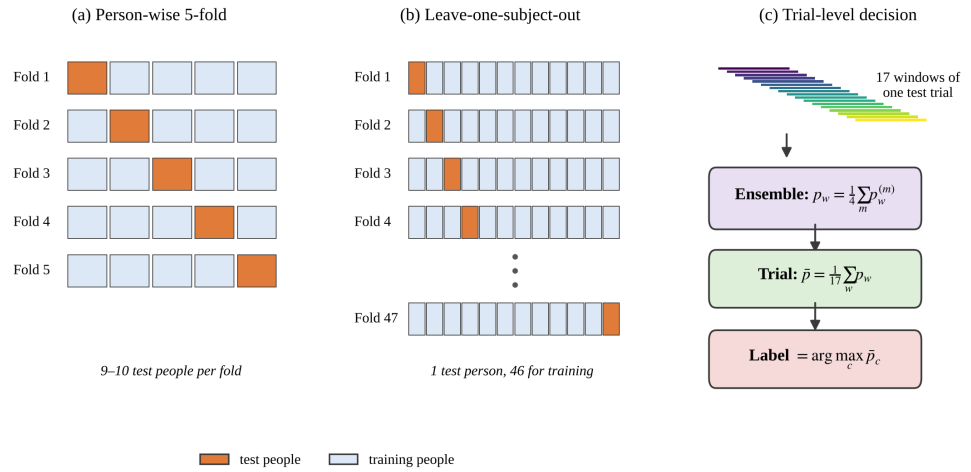


Fig. 10 Evaluation scheme: (a) person-wise 5-fold cross-validation; (b) leave-one-subject-out validation; (c) trial-level decision.

Validity checks. Several further checks were run on the final pipeline. (1) *Label permutation*: the labels were shuffled across trials within each recording and the complete pipeline, including alignment, mirror augmentation, the ensemble and trial averaging, was run again; this was done three times for every task, twenty more times for the binary task, and five times under LOSO (four runs for the binary task and one for the imagery-versus-imagery task). (2) *Reproducibility*: the pipelines were rerun with the same random seed, and the binary pipeline also with three other seeds. (3) *Learning curve*: for the binary task, the same test participants were classified by models trained on 5, 10, 20 and 30 randomly chosen training participants and on all of them. (4) *Nested cross-validation*: inside each outer training set, an inner person-wise cross-validation chose the configuration (no alignment, alignment, or alignment with mirror augmentation; for the imagery-versus-imagery task, standard or imagery-only calibration with or without mirroring) and the classifier, and only that choice was tested on the outer test participants. This removes the optimism that comes from comparing several set-ups on the same test people. (5) *Onset sensitivity*: every detected onset was shifted by ± 2 s and the classification was repeated.

Metrics. Accuracy is reported as the number of correctly classified test trials pooled over all folds, and as the mean $\pm$ standard deviation across folds. For the LOSO results we also report balanced accuracy, macro-averaged F1, Cohen's kappa, the accuracy per participant, and the accuracy on the training participants, which shows the gap between fit and generalisation. Chance level is 50% for the binary tasks, 16.67% for six classes and 8.33% for twelve classes. Accuracies were compared with chance using

a one-sided binomial test, and paired comparisons between two methods on the same trials used McNemar's test.

4 Results and Discussion

(to be added)

5 Conclusion

(to be added)

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